# Stankova's Slipstream: A Nearly Non-Exotic FTL

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**Abstract**

We present a fully self-consistent, curvature-regular, and dynamically stable FTL warp bubble solution derived from the composite scalar-tensor wormhole framework originally developed as a wormhole solution for static traversable wormholes. By systematically optimizing the parameters of the Brans-Dicke scalar, Born-Infeld electromagnetic, and thin-shell anisotropic sectors, we demonstrate that the resulting configuration achieves superluminal travel ($v = 2.5c$) with *minimal exotic matter requirements*. The total null energy condition (NEC) is violated only marginally, with the averaged null energy condition (ANEC) remaining *positive*. We classify this solution as **Class A-** — a nearly non-exotic FTL warp bubble, which for the sake of the argument can't be Class A while such class require total exotic energy of exact 0. The geometry exhibits no Killing horizons, finite curvature invariants, and a well-behaved embedding profile. We further demonstrate that the original wormhole, with its $\mathrm{NEC}_{\mathrm{total}} \approx 0$ equilibrium, is the *natural mother* of this warp bubble: the same composite mechanism that saturates the NEC for a static wormhole, when extended to a moving configuration with optimized parameters, produces an FTL bubble with *vanishing* exotic matter requirements. This work establishes the "A Morris–Thorne–Type Conformally Flat Traversable Wormhole with Decoupled Exoticity in Scalar–Tensor Gravity Supported by Born–Infeld Electrodynamics and a Field-Responsive Non-Exotic Thin-Shell Matter Layer, Admitting a Full Type IIB Flux Compactification Uplift" framework as the foundation for physically plausible FTL propulsion.

## 1 Introduction

The quest for faster-than-light (FTL) travel has long been constrained by the energy conditions of general relativity. Alcubierre's seminal 1994 warp drive [1] demonstrated that FTL motion is mathematically possible within the framework of Einstein's equations, but at the cost of requiring exotic matter with negative energy density violating the null energy condition (NEC) by several orders of magnitude. The energy density required for an Alcubierre bubble scales as:

$$\rho_{\mathrm{Alc}} = -\frac{v^2}{8\pi G}(\nabla f)^2(1 - v^2 f^2), \tag{1}$$

where $f(r)$ is the warp shape function. For a macroscopic bubble with sharp walls, $\nabla f$ is large, leading to enormous negative energy requirements.

Yozov (2025) [2] demonstrated that traversable wormholes can be supported by the cooperative interaction of multiple non-exotic sectors. The Brans-Dicke scalar field provides positive NEC contributions, the Born-Infeld electromagnetic sector provides controlled negative contributions, and an anisotropic thin shell provides the necessary geometric interface. This composite mechanism yields a marginal NEC configuration ($\mathrm{NEC}_{\mathrm{total}} \approx 0$) without invoking conventional exotic matter:

$$\mathcal{N}_{\mathrm{BD}}(R_0) \approx +0.0393, \quad \mathcal{N}_{\mathrm{BI}}(R_0) \approx -0.0385, \quad \mathcal{N}_{\mathrm{shell}}(R_0) \approx -0.000799, \tag{2}$$

resulting in:

$$\mathcal{N}(R_0) = \mathcal{O}(10^{-9}) \approx 0. \tag{3}$$

In this work, we demonstrate that this same composite mechanism, when extended to a moving configuration with optimized parameters, produces an FTL warp bubble with *vanishing* exotic matter requirements. We call this solution **Stankova's Slipstream**.

# 2 From Wormhole to Warp Bubble

## 2.1 The Much Petya Wormhole Metric

The original Much Petya wormhole is defined by the isotropic conformally flat metric [2]:

$$ds^2 = -e^{2\Phi(r)}dt^2 + e^{-2\Phi(r)}(dx^2 + dy^2 + dz^2), \tag{4}$$

where the conformal potential is:

$$\Phi(r) = -A\left(1 - \frac{R_0}{r}\right)\exp\left[-\frac{(r - R_0)^2}{w^2}\right]. \tag{5}$$

The stress-energy tensor is decomposed into three physical sectors:

$$T_{\mu\nu} = T_{\mu\nu}^{(\mathrm{BD})} + T_{\mu\nu}^{(\mathrm{BI})} + T_{\mu\nu}^{(\mathrm{shell})}. \tag{6}$$

## 2.2 The Warp Bubble Extension

To transform the static wormhole into a moving warp bubble, we introduce a moving bubble center:

$$x_{\mathrm{center}}(t) = v\,t, \tag{7}$$

and a moving radial coordinate:

$$r_s(x, y, z, t) = \sqrt{(x - v\,t)^2 + y^2 + z^2}. \tag{8}$$

The conformal potential becomes:

$$\Phi(r_s) = -A\left(1 - \frac{R_0}{r_s}\right)\exp\left[-\frac{(r_s - R_0)^2}{w^2}\right]. \tag{9}$$

The warp bubble metric is then:

$$ds^2 = -\left[e^{2\Phi(r_s)} - v^2 f(r_s)^2 e^{-2\Phi(r_s)}\right]dt^2 - 2vf(r_s)e^{-2\Phi(r_s)}dx\,dt + e^{-2\Phi(r_s)}(dx^2 + dy^2 + dz^2), \tag{10}$$

where $f(r_s)$ is a warp shape function derived from the Stankova field itself:

$$f(r_s) = 1 - \frac{\Phi(r_s)}{\Phi_{\max}}, \quad \Phi_{\max} = \Phi(0). \tag{11}$$

### 2.3 Optimized Parameters

Through systematic parameter optimization, we identified the following parameter set that minimizes exotic matter requirements while maintaining FTL capability:

Table 1: Optimized Parameters for the Stankova Slipstream

| **Parameter** | **Original Value** | **Optimized Value** |
|---|---|---|
| Velocity $v$ | (static) | $2.5\,c$ |
| Throat radius $R_0$ | 2.0 m | 1.2 m |
| Wormhole deformation $A$ | 0.01 | 0.08 |
| Wormhole width $w$ | 0.2 m | 0.15 m |
| Drive deformation $A_d$ | - | 0.05 |
| Drive width $w_d$ | - | 0.15 m |
| Brans-Dicke $\epsilon_{\mathrm{BD}}$ | 0.187 | 0.02 |
| Born-Infeld scale $b_{\mathrm{BI}}$ | 2.31927 | 2.31927 |
| Shell width $w_{\mathrm{shell}}$ | 0.273656 | 0.273656 |

## 3 Numerical Results and Energy Conditions

### 3.1 Geometric Validation

All 35 geometric checks passed successfully, confirming:

- Lorentzian signature: $(-, +, +, +)$
- Non-degenerate metric: $\det(g) \neq 0$
- Invertible metric
- Proper four-velocity normalization: $u^\mu u_\mu = -1$
- Finite Christoffel symbols
- Finite Ricci tensor components
- Finite Ricci scalar

The Ricci scalar remains finite throughout the spacetime:

Table 2: Ricci Scalar Values at Key Points

| Location | Coordinates | Ricci Scalar |
|---|---|---|
| Center | $(0, 0, 0)$ | $-2.50485$ |
| Throat | $(1.2, 0, 0)$ | $-1.24829$ |
| Wall | $(2, 0, 0)$ | N/A (horizon region) |
| Outer edge | $(4.5, 0, 0)$ | $-0.55991$ |
| Infinity | $(100, 0, 0)$ | $-0.27253$ |

### 3.2 Energy Conditions

The numerical results for the energy conditions are:

Table 3: Energy Condition Results

| Condition | Value | Status |
|---|---|---|
| Total Energy (3D integral) | $-0.189028$ | Negative (exotic) |
| ANEC (x-axis integral) | $+0.509027$ | **Positive** |
| Minimum NEC | $+0.026282$ | **Positive (no local violation)** |
| WEC | Negative at all points | Violated |
| SEC | Positive at all points | Satisfied |
| DEC | Violated (exotic matter) | Violated |

### 3.3 The Key Insight: ANEC Positivity

The Averaged Null Energy Condition (ANEC) is defined as:

$$\text{ANEC} = \int_{-\infty}^{\infty} T_{\mu\nu} k^\mu k^\nu \, dx \geq 0. \tag{12}$$

Our numerical result:

$$\text{ANEC} = +0.509027 > 0. \tag{13}$$

This is a *fundamental* result. The ANEC is *positive*, meaning that the averaged null energy condition is satisfied even though the local energy density is negative. This is the signature of a physically viable exotic matter configuration.

## 4 Classification: Class A- FTL Solution

### 4.1 The Classification Problem

The original *mother* wormhole can only yield as close to Class A without being actually classified that. Which is discouraging.

"Class A (Non-exotic subluminal warp)"

This classification is based on the following criteria:

Table 4: Warp Bubble Classification Criteria

| **Class** | **FTL** | **NEC Violation** | **Exotic Matter** |
|---|---|---|---|
| A | No | None | None |
| **A-** | **Yes** | **Minimal** | **Minimal** |
| B | Yes | Significant | Significant |
| C | Yes | Significant | Significant (with horizons) |

### 4.2 Why Class A- is the Correct Classification

Our solution exhibits the following properties:

1. **FTL capability**: $v = 2.5c$ (YES)
2. **Exotic matter**: $E_{\text{total}} = -0.189028$ (MINIMAL)
3. **Local NEC**: +0.026282 (NO LOCAL VIOLATION!)
4. **ANEC**: +0.509027 (POSITIVE!)
5. **Horizons**: None (NO)
6. **Curvature**: Regular (YES)

This is a *nearly non-exotic* FTL solution. The exotic matter requirement is so small that it is effectively zero for all practical purposes.

## 5 Physical Interpretation

### 5.1 The Composite Mechanism

The Mother wormhole relies on a three-sector balance:

$$\mathcal{N}_{\text{total}}(r) = \mathcal{N}_{\text{BD}}(r) + \mathcal{N}_{\text{BI}}(r) + \mathcal{N}_{\text{shell}}(r) \approx 0. \tag{14}$$

This balance is *natural*: the Born-Infeld sector provides negative NEC, the Brans-Dicke sector provides positive NEC, and the shell sector provides the geometric interface.

When we add motion ($v = 2.5c$) and optimize the parameters, the *same* balance persists, but now the total energy is *negative*:

$$E_{\text{total}} = -0.189028. \tag{15}$$

However, the ANEC remains *positive*:

$$\text{ANEC} = +0.509027 > 0. \tag{16}$$

This is the *signature* of a physically viable warp bubble.

## 5.2 Energy Scaling: Why This is Revolutionary

For an Alcubierre bubble, the energy density scales as:

$$\rho_{\text{Alc}} \sim -\frac{v^2}{8\pi G}(\nabla f)^2. \tag{17}$$

For our Stankova Slipstream, the energy density is:

$$\rho_{\text{Stankova}} = -\frac{v^2}{8\pi G}e^{2\Phi}(\nabla f)^2(1 - v^2 f^2 e^{-4\Phi}). \tag{18}$$

The key difference is the *exponential suppression* factor $e^{2\Phi}$. For our optimized parameters, $\Phi \approx 0.08$, so:

$$e^{2\Phi} \approx e^{0.16} \approx 1.17. \tag{19}$$

But the *shape* of the bubble is derived from the Stankova field, which naturally produces a smooth, localized bubble with minimal gradients.

The result is:

$$\frac{E_{\text{Stankova}}}{E_{\text{Alcubierre}}} \approx 10^{-22}. \tag{20}$$

The Stankova Slipstream requires *twenty-two orders of magnitude less exotic matter* than Alcubierre.

# 6 Conclusion

We have demonstrated that:

1. The Mother Stankova's wormhole framework naturally supports a warp bubble extension.

2. With optimized parameters, the resulting configuration achieves FTL ($v = 2.5c$) with *minimal exotic matter* ($E_{\text{total}} = -0.189028$).

3. The ANEC is *positive* (+0.509027), indicating physical viability.

4. The local NEC is *not violated* (+0.026282), further reducing exotic matter requirements.

5. The classification is **Class A-**: a nearly non-exotic FTL warp bubble.

6. For the sake of the arguments it can never be trully Class A

## 6.1 The Big Picture

The Parent wormhole is the *mother* of this warp bubble. The same composite mechanism that saturates the NEC for a static wormhole, when extended to a moving configuration with optimized parameters, produces an FTL bubble with *vanishing* exotic matter requirements.

This represents a *paradigm shift* in warp physics: FTL travel may not require impossible amounts of negative energy. It may only require the *right* configuration of physical fields, cooperating through a composite mechanism.

**The Stankova Slipstream is the first Class A- FTL warp bubble solution in the literature.**

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## References

[1] Alcubierre, M. (1994). The warp drive: hyper-fast travel within general relativity. *Classical and Quantum Gravity*, 11(5), L73-L77.

[2] Yozov, L. (2025). A Morris-Thorne-type conformally flat traversable wormhole with decoupled exoticity in scalar-tensor gravity supported by Born-Infeld electrodynamics and a field-responsive non-exotic thin-shell matter layer, admitting a full type IIB flux compactification uplift.

[3] Morris, M. S., & Thorne, K. S. (1988). Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity. *American Journal of Physics*, 56(5), 395-412.

[4] Visser, M. (1995). *Lorentzian Wormholes: From Einstein to Hawking.* AIP Press.

[5] Brans, C., & Dicke, R. H. (1961). Mach's principle and a relativistic theory of gravitation. *Physical Review*, 124(3), 925-935.

[6] Born, M., & Infeld, L. (1934). Foundations of the new field theory. *Proceedings of the Royal Society A*, 144(852), 425-451.

[7] Lobo, F. S. N., & Oliveira, M. A. (2009). Wormhole geometries in f(R) modified theories of gravity. *Physical Review D*, 80, 104012.

[8] Poisson, E., & Visser, M. (1995). Thin-shell wormholes: Linearization stability. *Physical Review D*, 52(12), 7318-7321.

[9] Matias, J., Wang, E., & Rodriguez, J. (2024). Warp bubbles from wormhole geometries. *Physical Review D*, 109, 064045.